Coalescence and drag in bubbly channel flows

Annual Meeting of the Japanese Society of Fluid Mechanics,
2021 September 21

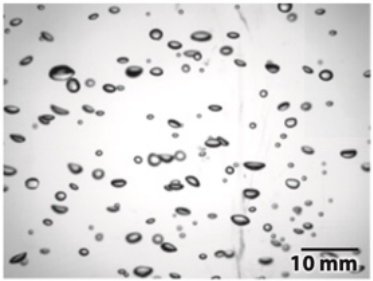
Ianto Cannon, supervised by Marco E. Rosti,
Complex Fluids and Flows unit



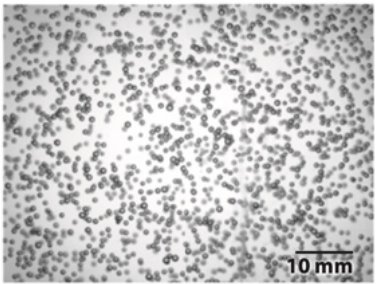
OIST

OKINAWA INSTITUTE OF SCIENCE AND TECHNOLOGY GRADUATE UNIVERSITY
沖縄科学技術大学院大学

"The **coalescence** of small bubbles hardly occurs in the presence of surfactant"[1].



Bubbles in tap water.



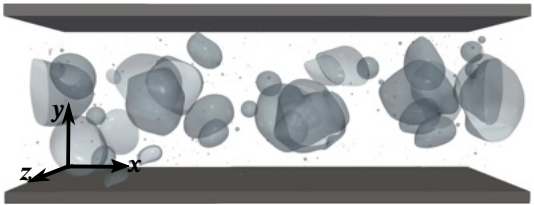
Bubbles in tap water with 2ppm Triton X-100 solution.

[1] Takagi & Matsumoto 2011 *Annual Review of Fluid Mechanics* 43, 615–636

Methods

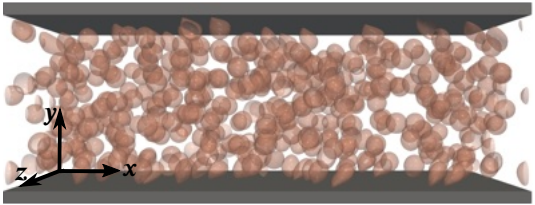
In the **volume of fluid** (VOF) method, an indicator function H is advected with the fluid,[2]

$$\frac{\partial H}{\partial t} + \mathbf{u} \cdot \nabla H = 0. \tag{1}$$



In the **front tracking** (FT) method, the velocity of the interface is a sum over fluid velocities at nearby grid points,[3]

$$\mathbf{u}^{(l)} = \sum_i D(\mathbf{x}_i - \mathbf{x}^{(l)})\mathbf{u}_i. \tag{2}$$

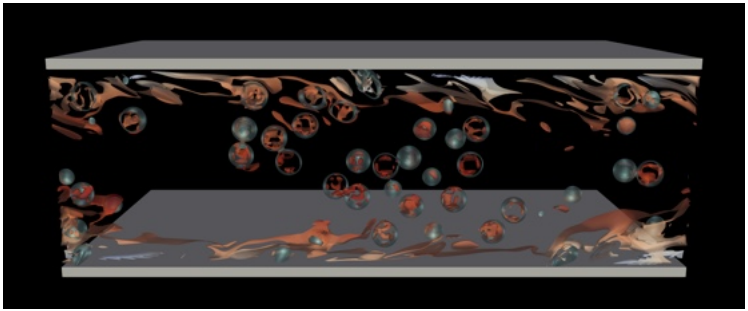


[2] Unverdi & Tryggvason 1992 *Journal of Computational Physics* 100, 25–37.

[3] Ii, Sugiyama, Takeuchi, Takagi, Matsumoto & Xiao 2012 *Journal of Computational Physics* 231, 2328–2358.

Setup: pressure driven channel flow

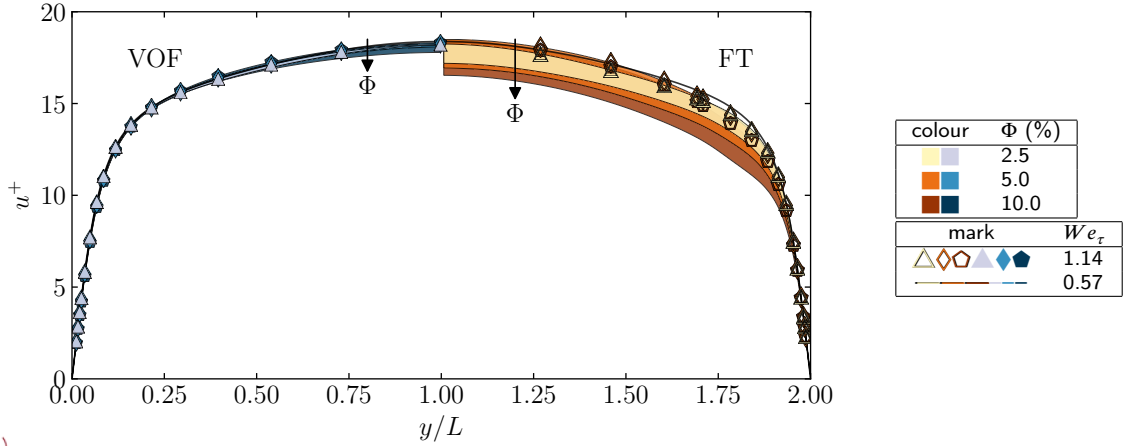
Channel dimensions	$6L \times 2L \times 3L$
VOF: uniform grid	$1296 \times 432 \times 648$
FT: non uniform grid	$566 \times 240 \times 288$



Friction Reynolds number	$Re_\tau \equiv u_\tau L/\nu = 180$
Four volume fractions	$\Phi = 0\%, 2.5\%, 5\%, 10\%,$
Two values of surface tension	$We_\tau \equiv \rho_c u_\tau^2 L/\gamma = 0.57, 1.14$

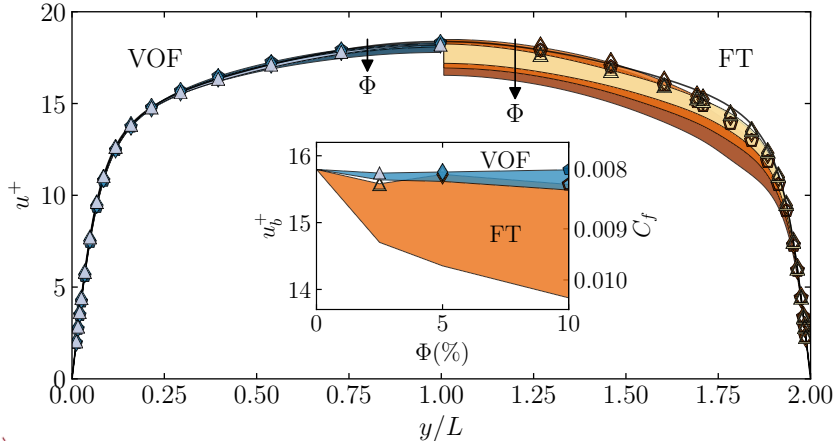
Flow profile

As the volume fraction of bubbles increases, the **flow rate is reduced**.
Non-coalescing bubbles have a much greater effect.



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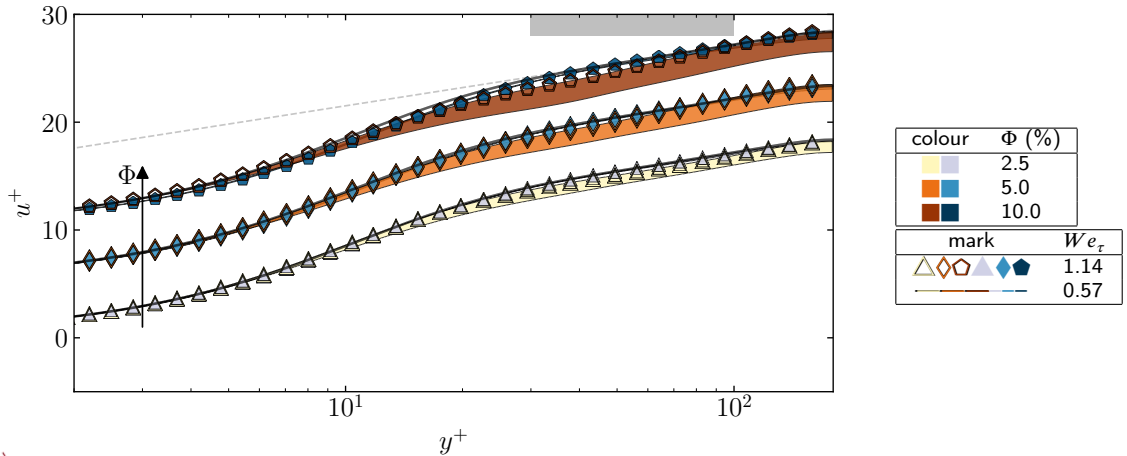


colour	Φ (%)
	2.5
	5.0
	10.0

mark	We_τ
	1.14
	0.57

Log-law

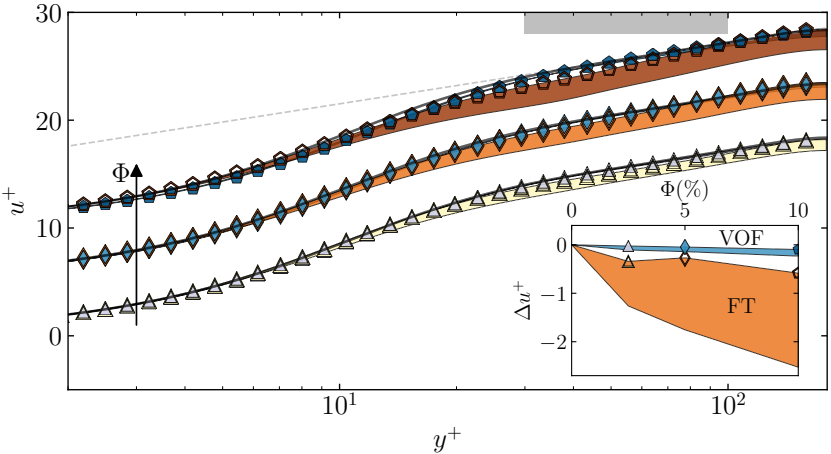
The streamwise velocity profiles follow the log-law for a turbulent channel flows, with a **negative shift**[4] $u^+ = 2.5 \log y^+ + 5.5 + \Delta u^+$.



[4] von Kármán, T. 1930. Mechanische Ahnlichkeit und Turbulenz. Math-Phys. Klasse.

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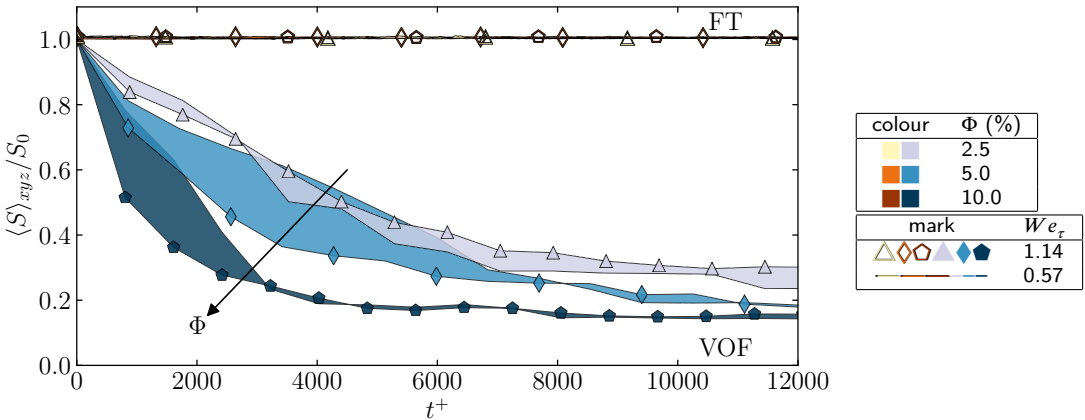
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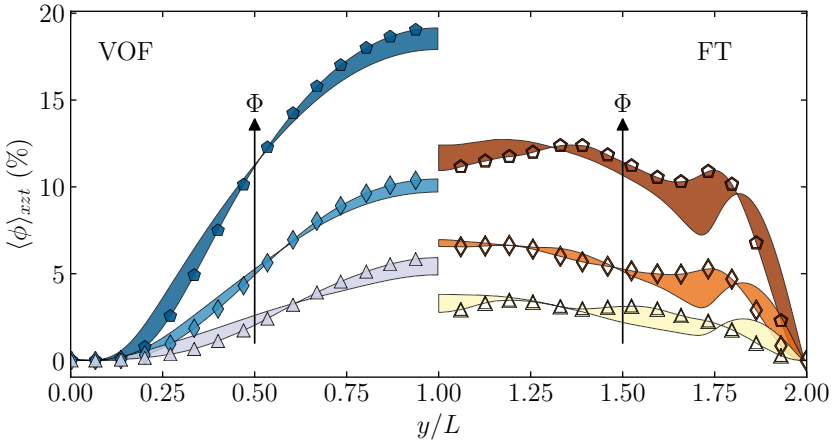
[4] von Kármán, T. 1930. Mechanische Ahnlichkeit und Turbulenz. Math-Phys. Klasse.

Total area of the bubbles

Coalescence leads to a significant **decrease in surface area**.

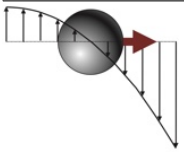


Volume fraction of dispersed phase



colour	Φ (%)
	2.5
	5.0
	10.0

mark	We_τ
	1.14
	0.57



Martel & Toner 2014 Ann. Rev. of Bio. Eng. 16, 371–396

Volume of fluid bubbles are able to coalesce and move to a region of low shear[5].

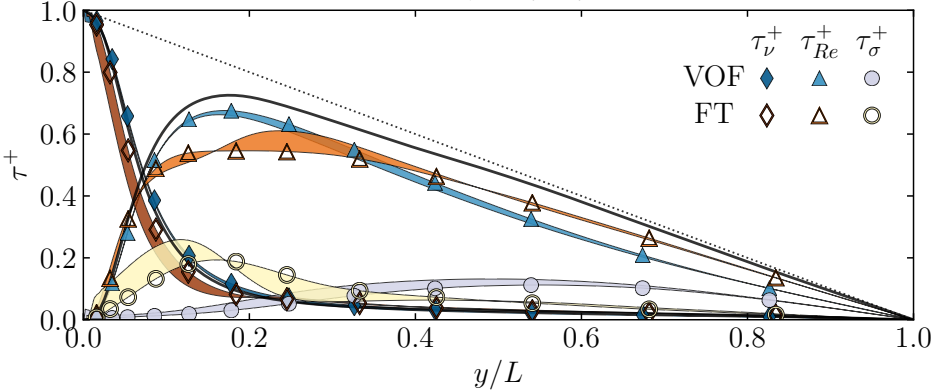
Front tracking bubbles experience a **shear-gradient lift force**, pulling them toward the wall.

[5] Alghalibi, Rosti & Brandt 2019 Phy. Rev. Fluids, 4(10):104201

Shear stresses

The shear stress is a sum of the viscous, inertial, and surface tension parts,

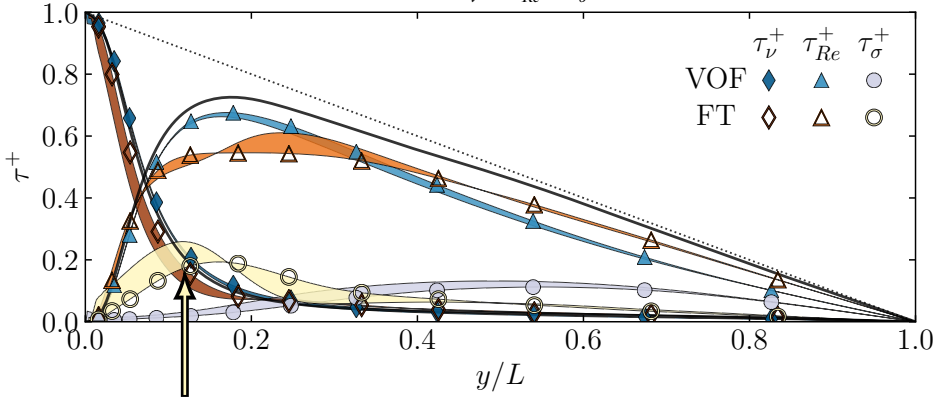
$$1 - y/L = \tau_\nu^+ + \tau_{Re}^+ + \tau_\sigma^+ \tag{3}$$



Shear stresses

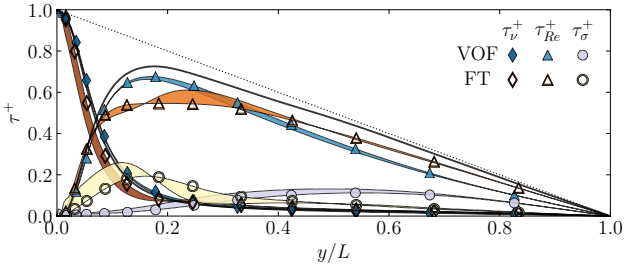
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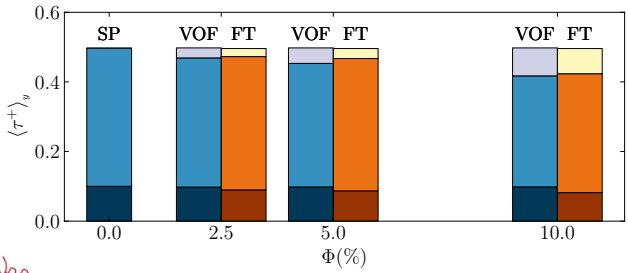
$$1 - y/L = \tau_\nu^+ + \tau_{Re}^+ + \tau_\sigma^+ \tag{3}$$



The front tracking **surface tension stress** has a peak near the wall.

Shear stresses



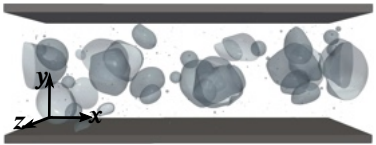


The mean viscous shear stress is reduced in the front tracking simulations.
This stress is proportional to the centreline velocity,

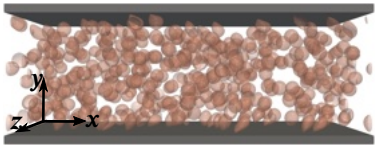
$$\langle \tau_\nu^+ \rangle_y = \frac{1}{L} \int_0^L \frac{du^+}{dy^+} dy = \frac{\delta_\nu}{L} u_{cen}^+. \quad (4)$$

Hence the flow velocity is reduced in the front tracking simulations.

Conclusions & future work



Volume of fluid method

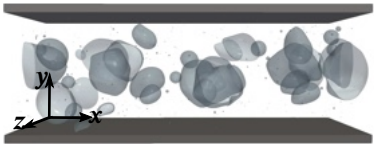


Front tracking method

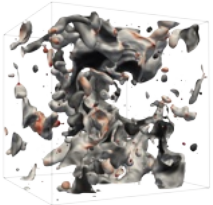
1. Non-coalescing bubbles produce a significant **drag increase** compared to coalescing bubbles.
2. The non-coalescing bubbles experience a **shear-gradient lift force**, which attracts them to the viscous sublayer in the channel.
3. This leads to large surface tension stresses near the channel wall, **reducing the budget for viscous stress**, and increasing drag.

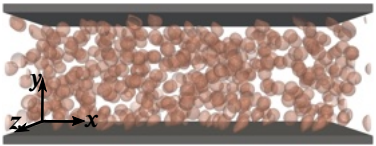
I. Cannon, D. Izbassarov, O. Tammisola, L. Brandt, and **M. E. Rosti**, *The effect of droplet coalescence on drag in turbulent channel flows*, Physics of Fluids, 33.8, p85112, Aug. 2021, 10.1063/5.0058632.

Conclusions & future work



Volume of fluid method


Surfactant model



Front tracking method

1. Non-coalescing bubbles produce a significant **drag increase** compared to coalescing bubbles.
2. The non-coalescing bubbles experience a **shear-gradient lift force**, which attracts them to the viscous sublayer in the channel.
3. This leads to large surface tension stresses near the channel wall, **reducing the budget for viscous stress**, and increasing drag.
4. Future work with **G. Soligo**: modelling the surfactant as a **third fluid phase**

Thank you! Questions?

I. Cannon, D. Izbassarov, O. Tammisola, L. Brandt, and **M. E. Rosti**, *The effect of droplet coalescence on drag in turbulent channel flows*, Physics of Fluids, 33.8, p85112, Aug. 2021, 10.1063/5.0058632.